

## Production &amp; Costs

2 Goods

Supply

Input demands

Production:

Combining inputs to make output.

1. Getting the most possible output from a set of inputs. (Production efficiency)
2. Find the input mix that gets a level of output as cheaply as possible.

Inputs are classified as fixed or variable.

Fixed — can't <sup>be</sup> changed during the time frameyou are considering " doesn't change with output ".

variable — can be changed during the relevant time frame. " Change with output ".

actually opportunity cost

Short run: A time frame is considered short-run if at least one input remains fixed. (operation)

Long-run: Minimum amount of time it takes to change all input. (planning)

Costs: What you give up to use an input.

Fixed cost: Cost of all of the fixed inputs.  
FC

Variable cost: Cost of the variable inputs.  
VC

Total cost: Cost of all your inputs.  
TC

$$TC = FC + VC$$

Unavoidable fixed cost — license



⑦ Cost minimization subject to a technology constraint

$$q = \frac{10KL}{K+L}$$

Capital: an input (used repeatedly) to make something else.

└ Diminishing returns due to depreciation →  $\delta \downarrow$  or  $c \uparrow$   
 Increasing  $\delta$  due to innovation. └

$$\bar{K} = 1000$$

$$q_{SR} = \frac{1000L}{100+L}$$

SR → short run

$$\Rightarrow 1000L = 100q + qL$$

$$\Rightarrow 1000L - qL = 100q$$

$$\Rightarrow L(1000 - q) = 100q$$

$$\Rightarrow L^* = \frac{100q}{1000 - q} \quad \text{for } (1000 - q) > 0$$

$$q = \frac{100KL}{K+L}$$

$$\Rightarrow MP_L = \frac{\partial q}{\partial L}$$

$$= \frac{1000(100+L) - 1000L}{(K+L)^2 (100+L)^2}$$

$$= \frac{100000 + 1000L - 1000L}{(100+L)^2}$$

$$MP_L^{SR} = \frac{100000}{(100+L)^2}$$

$$q = \frac{10KL}{K+L}$$

$$MP_K = \frac{\partial q}{\partial K}$$

$$= \frac{(K+L) 10L - 10KL}{(K+L)^2}$$

$$MP_L = \frac{(K+L) 10K - 10KL}{(K+L)^2}$$



$$\frac{MP_L}{MP_K} = MRTS$$

$$\Rightarrow \frac{\cancel{10K^\gamma} + \cancel{10KL^\gamma} - \cancel{10KE}}{\cancel{10K^\gamma} + \cancel{10L^\gamma} - \cancel{10KE}}$$

$$\Rightarrow MRTS = \frac{10K^\gamma / (K+L)^\gamma}{10L^\gamma / (K+L)^\gamma}$$

$$= \frac{10K^\gamma}{(K+L)^\gamma} \cdot \frac{(K+L)^\gamma}{10L^\gamma}$$

$$= \frac{K^\gamma}{L^\gamma}$$

$$= \left(\frac{K}{L}\right)^\gamma$$

$K \rightarrow$  compensate  $- v$

$L \rightarrow$  wage  $- w$

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$$\text{Min}_{L, K} \quad \omega L + vK = c(q, \omega)$$

$$\text{s.t.} \quad q^* = f(K, L) \Rightarrow q = \frac{10KL}{K+L}$$

$$\text{Min}_{L, K, \lambda} \quad \mathcal{L} = \omega L + vK + \lambda \left[ q - \frac{10KL}{K+L} \right]$$

$$\frac{\partial \mathcal{L}}{\partial L} = \omega - \lambda \left[ \frac{10K^2}{(K+L)^2} \right] = 0 \quad \text{--- (I)}$$

$$\frac{\partial \mathcal{L}}{\partial K} = v - \lambda \left[ \frac{10L^2}{(K+L)^2} \right] = 0 \quad \text{--- (II)}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = q - \frac{10KL}{K+L} = 0 \quad \text{--- (III)}$$

$$\Rightarrow q = \frac{10KL}{K+L}$$

$$\omega = \lambda \left[ \frac{10K^2}{(K+L)^2} \right]$$

$$v = \lambda \left[ \frac{10L^2}{(K+L)^2} \right]$$

from ① & ②

$$\frac{w}{v} = \frac{\lambda \frac{10k^r}{(k+l)^r}}{\lambda \cdot \frac{10L^r}{(k+l)^r}}$$

$$= \frac{k^r}{L^r}$$

$$\therefore \boxed{\frac{w}{v} = \frac{k^r}{L^r}} = \text{MRTS} \quad \text{--- (iv)}$$

$$\Rightarrow k^r v = w L^r$$

$$\Rightarrow k^r = \frac{w L^r}{v}$$

$$\Rightarrow k = \sqrt{\frac{w}{v}} \cdot L \quad \text{--- (v)}$$

$$\& L = \sqrt{\frac{v}{w}} \cdot K \quad \text{--- (vi)}$$

Substituting (v) into (iii)

$$q = \frac{10 L \cdot \sqrt{\frac{w}{v}} \cdot L}{\sqrt{\frac{w}{v}} \cdot L + L}$$

$$= \frac{L (10 L \sqrt{\frac{w}{v}})}{L (\sqrt{\frac{w}{v}} + 1)}$$



$$\therefore q = \frac{10L \sqrt{\frac{\omega}{v}}}{\left(\sqrt{\frac{\omega}{v}} + 1\right)}$$

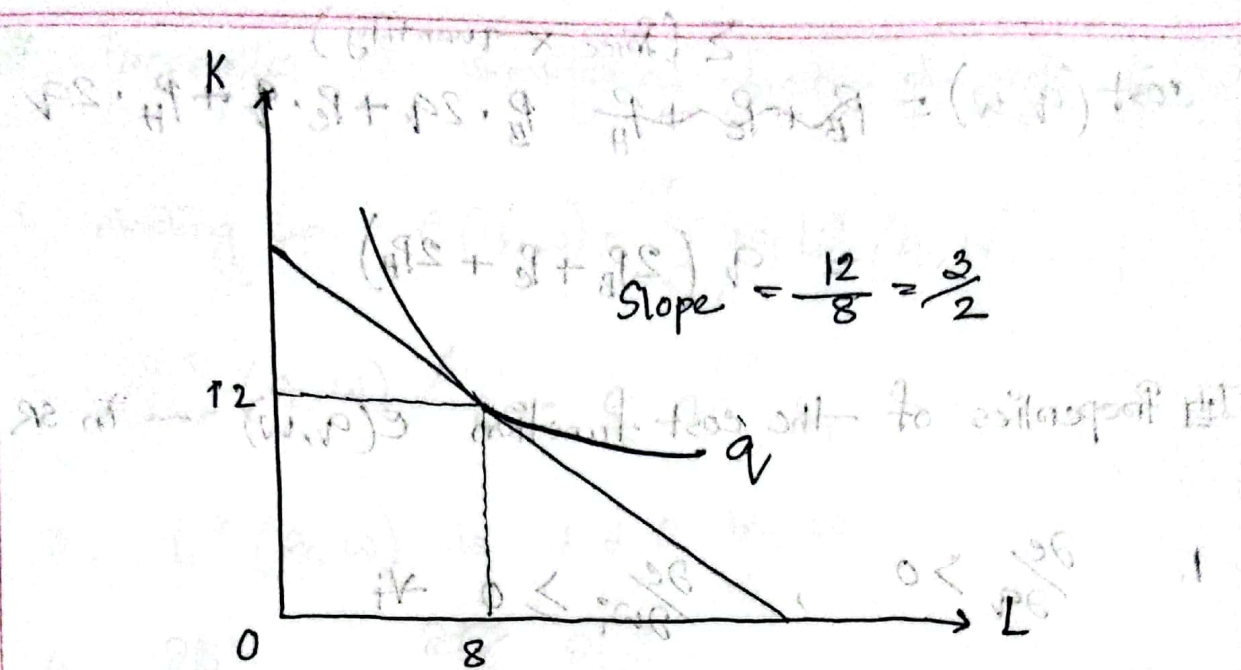
$$\Rightarrow 10L \sqrt{\frac{\omega}{v}} = q \left(\sqrt{\frac{\omega}{v}} + 1\right)$$

$$\Rightarrow L^* = \frac{q \left(\sqrt{\frac{\omega}{v}} + 1\right)}{10 \sqrt{\frac{\omega}{v}}}$$

$$\Rightarrow K = \sqrt{\frac{\omega}{v}} \cdot L$$

$$= \sqrt{\frac{\omega}{v}} \cdot \frac{q \left(\sqrt{\frac{\omega}{v}} + 1\right)}{10 \sqrt{\frac{\omega}{v}}}$$

$$= \frac{q \left(\sqrt{\frac{\omega}{v}} + 1\right)}{10}$$



Leontief / fixed proportion:

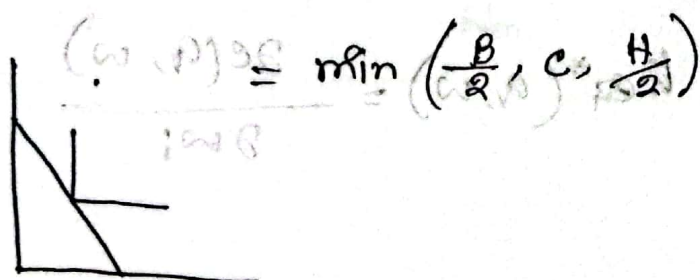
Simple example: Sandwich

2 slices of Bread (B)

1 slice of cheese (C)

2 slices of Ham (H)

$q = \min\left(\frac{B}{2}, C, \frac{H}{2}\right)$  is the number of Sandwiches



$$\Rightarrow B = 2q$$

$$\frac{B}{2} = C = \frac{H}{2}$$

as  $q = C$  &  $q = \frac{H}{2}$



$\Sigma (\text{Price} \times \text{quantity})$

$$\text{cost}(q, w) = \cancel{P_B + P_C + P_H} \quad P_B \cdot 2q + P_C \cdot q + P_H \cdot 2q$$

$$= q(2P_B + P_C + 2P_H)$$

Properties of the cost function  $c(q, w)$  — In SR

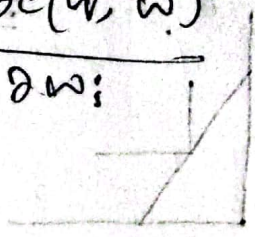
$$1. \quad \frac{\partial c}{\partial q} > 0, \quad \frac{\partial c}{\partial w_i} \geq 0 \quad \forall i$$

$\frac{\partial c}{\partial w_j} > 0$  for at least one  $j$   
 : non-proportional input-output

2. H.d.1 in  $w$
3. concave in  $w$
4.  $c(q, w)$  is continuously twice differentiable except at possibly a countable number of points.

5. Shephard's Lemma

$$L^*(q, w) = \frac{\partial c(q, w)}{\partial w_i}$$

$$l_i^*(q, w) = \frac{\partial c(q, w)}{\partial w_i}$$




## Properties of conditional factor demands $l_i^*(q, w)$

1. Adding up  $C(q, w) = \sum_{i=1}^n w_i l_i^*(q, w)$

2.  $\frac{\partial l_i^*(q, w)}{\partial w_i} \leq 0 \quad \forall i$

3.  $l_i^*(q, w)$  is h.d.o in  $w$

4.  $\frac{\partial l_i^*}{\partial w_k} = \frac{\partial l_k^*}{\partial w_i}$   $w_k \rightarrow$  price of capital

II

$$q = (KLM)^{\frac{1}{3}}$$

$$\frac{w_m}{w} = \text{price of materials}$$

Short run,  $K = 64$

$$q_{SR} = 4(LM)^{\frac{1}{3}}$$

Tangency:  $\frac{M}{L} = \frac{w}{w_m}$

$$M = \frac{Lw}{w_m}$$

① in ①

$$q = 4 \left( L \cdot L \cdot \frac{W}{W_m} \right)^{\frac{1}{3}}$$

$$\Rightarrow q^3 = 64 \left( L^2 \cdot \frac{W}{W_m} \right)$$

$$\Rightarrow \frac{q^3}{64} = L^2 \cdot \frac{W}{W_m}$$

$$\Rightarrow L^2 = \frac{q^3}{64} \cdot \frac{W_m}{W}$$

$$\Rightarrow L^* = \frac{q^{\frac{3}{2}}}{8} \cdot \sqrt{\frac{W_m}{W}}$$

Now, ① in ②

$$M = \frac{q^{\frac{3}{2}}}{8} \sqrt{\frac{W_m}{W}} \cdot \frac{W}{W_m}$$

$$\Rightarrow M^* = \frac{q^{\frac{3}{2}}}{8} \cdot \sqrt{\frac{W}{W_m}}$$

$$e(q, W) = 64V + \frac{q^{\frac{3}{2}}}{8} \sqrt{W_m \cdot W} + \frac{q^{\frac{3}{2}}}{8} \sqrt{W \cdot W_m}$$

$$= 64V + W \cdot \frac{q^{\frac{3}{2}}}{8} \sqrt{\frac{W_m}{W}} + W_m \frac{q^{\frac{3}{2}}}{8} \sqrt{\frac{W}{W_m}}$$

Q An individual's demand for chicken is given by  
 $Q = 150 p^{-2} p_{p_1}^{-0.5} p_{p_2}^x I^{0.8}$  where  $p$  = Price of chicken

$p_{p_1}$  = Price of related good 1

$p_{p_2}$  = " " " " 2

$I$  = Income in thousand.

a) Is this luxury, a necessity or an inferior good?

b) The two related goods are beef (a substitute for chicken) and rice (a complement for chicken).

Given what you know about elasticities, Good 1 or Good 2  
 is beef and Good 1 or Good 2 is rice.

c) The price of chicken increases. Will this person's expenditure on chicken increase, decrease or remain the same?

d) What is the value of  $x$ ?



$$\log Q = -2 \log P - 0.5 \log P_1 + 2 \log P_2 + 0.8 \log I$$

also > 0.

a)  $E_{Q,I} = \frac{d \log Q}{d \log I} = 0.8 < 1$  necessity.

b) Beef is  $P_2$ , rice  $P_1$

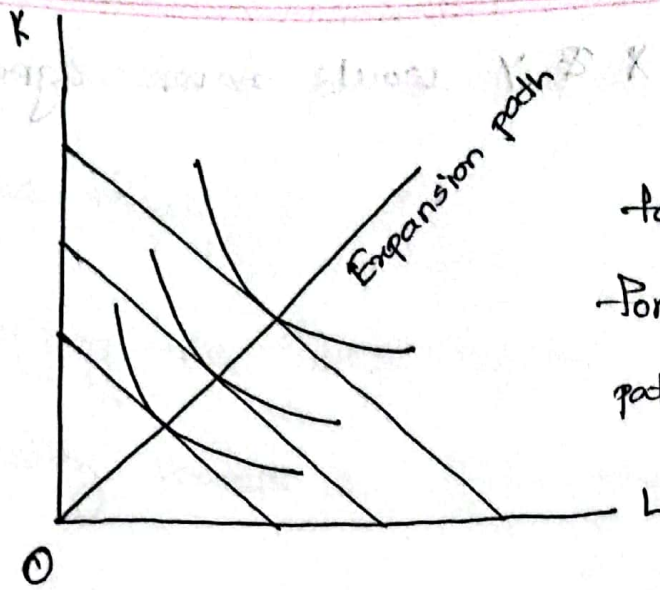
c)  $\frac{d \log Q}{d \log P} = (E_{Q,P}) = -2 \rightarrow$  elastic expenditure?

d) Sum of the elasticities of price and income are zero.

$$E_{Q,P} + E_{Q,P_1} + E_{Q,P_2} + E_{Q,I} = 0$$

$$\Rightarrow -2 - 0.5 + \alpha + 0.8 = 0$$

$$\Rightarrow \alpha = 1.7$$



when you solve tangency

for  $K$ , that is the formula for the expansion path.

Our example,

$$MRTS = \left(\frac{K}{L}\right)^2$$

$$\left(\frac{K}{L}\right)^2 = \frac{w}{v}$$

$$K = L \cdot \sqrt{\frac{w}{v}}$$

If  $f(K, L)$  is homothetic, then the expansion path is linear.

$$\begin{array}{ll} \text{Min} & wL + vK \\ \text{K, L} & \end{array}$$

$$\text{s.t. } q = f(K, L)$$

The value of elasticity depends on how easily one factor can be substituted for the other.

$\sigma \rightarrow$  bigger ; Easier it is to substitute one input for another.

Elasticity of substitution :  $\frac{\% \Delta \text{ in factor prices}}{\% \Delta \text{ in price of factors}}$

$$\sigma = \frac{\% \Delta \text{ in } \frac{K}{L}}{\% \Delta \text{ in } \left[ \frac{w}{r} \right]} \rightarrow \text{MRTS} = \frac{\% \Delta \text{ in } \frac{K}{L}}{\% \Delta \text{ in MRTS}_{L,K}}$$

$$C_{SR}(w, q) = 64v + \frac{q^{3/2}}{4} \sqrt{w_m w}$$

$$AC = \text{Average cost} = \frac{C(w, q)}{q}$$

$$= \frac{64v}{q} + \frac{q^{3/2}}{4q} \sqrt{w_m \cdot w}$$

$$= \frac{64v}{q} + \frac{q^{3/2-1}}{4} \sqrt{w_m \cdot w}$$

$$AC_{SR}(w, q) = \frac{64v}{q} + \frac{q^{1/2}}{4} \sqrt{w_m \cdot w}$$

$$\left( \frac{1}{q} \right) \cdot \frac{1}{q} = \frac{1}{q^2} \quad \left( \frac{1}{q} \right) \cdot \frac{1}{q} = \frac{1}{q^2}$$

$$\left( \frac{1}{q} \right) \cdot \frac{1}{q} = \frac{1}{q^2} \quad \left( \frac{1}{q} \right) \cdot \frac{1}{q} = \frac{1}{q^2}$$



$$MC = \text{marginal cost} = \frac{\partial c(q, w)}{\partial q} = \frac{\frac{3}{2} q^{\frac{3}{2}-1} \cdot \sqrt{w_m \cdot w}}{4}$$

$$MC_{SR} = \frac{3}{8} q^{\frac{1}{2}} \cdot \sqrt{w_m \cdot w}$$

□ Long-run solution with Lagrangian, as all factors of production is variable.

What about long-run

Now we can change  $K, L$  and  $M$ .

$$\min_{L, K, M, \lambda} \mathcal{L} = wL + vK + w_m M + \lambda [q - (KLM)^{\frac{1}{3}}]$$

$$\frac{\partial \mathcal{L}}{\partial L} = w - \frac{1}{3} \lambda \left( \frac{KM}{L^2} \right)^{\frac{1}{3}} = 0 \Rightarrow w = \frac{1}{3} \lambda \left( \frac{KM}{L^2} \right)^{\frac{1}{3}}$$

$$\frac{\partial \mathcal{L}}{\partial K} = v - \frac{1}{3} \lambda \left( \frac{LM}{K^2} \right)^{\frac{1}{3}} = 0 \Rightarrow v = \frac{1}{3} \lambda \left( \frac{LM}{K^2} \right)^{\frac{1}{3}}$$

$$\frac{\partial \mathcal{L}}{\partial M} = w_m - \frac{1}{3} \lambda \left( \frac{KL}{M^2} \right)^{\frac{1}{3}} = 0 \Rightarrow w_m = \frac{1}{3} \lambda \left( \frac{KL}{M^2} \right)^{\frac{1}{3}}$$

$$\frac{\partial \Delta}{\partial \lambda} = q - (KLM)^{\frac{1}{3}} = 0 \quad \text{--- (IV)}$$

from (I) & (II)

$$\frac{\omega}{v} = \frac{\frac{1}{2} \cdot \lambda \cdot K^{\frac{1}{3}} M^{\frac{1}{3}} L^{-\frac{2}{3}}}{\frac{1}{2} \cdot \lambda \cdot L^{\frac{1}{3}} M^{\frac{1}{3}} K^{-\frac{2}{3}}}$$

$$\Rightarrow \frac{\omega}{v} = K^{\frac{1}{3} + \frac{2}{3}} \cdot L^{-\frac{2}{3} - \frac{1}{3}} \quad \text{--- (V)}$$

$$\Rightarrow \boxed{\frac{\omega}{v} = \frac{K}{L}} \quad \text{--- (V)}$$

$$\Rightarrow K = L \cdot \frac{\omega}{v}$$

$$\therefore \boxed{\frac{M}{L} = \frac{\omega}{\omega_m}} \quad \text{--- (VI)}$$

$$\Rightarrow M = L \cdot \frac{\omega}{\omega_m}$$

in (IV)

$$q = (KLM)^{\frac{1}{3}}$$

$$= \left( \frac{\omega}{v} \cdot L \cdot L \cdot \frac{\omega}{\omega_m} \cdot L \right)^{\frac{1}{3}}$$

$$\Rightarrow q^3 = \frac{\omega}{v} \cdot \frac{\omega}{\omega_m} \cdot L^3 \Rightarrow L^3 = q^3 \cdot \frac{\omega_m v}{\omega^r}$$

$$\Rightarrow L^3 = L^* = q \cdot \left( \frac{\omega_m \cdot v}{\omega^r} \right)^{\frac{1}{3}} \quad \text{--- (vii)}$$

$$\Rightarrow \left( \frac{\omega^r}{\omega_m v} \right)^{\frac{1}{3}} = \frac{1}{L^*}$$

$$\text{(vii)} \quad q_m \quad \text{(v)}$$

$$K^* = q \left( \frac{\omega_m \cdot \omega}{v^r} \right)^{\frac{1}{3}}$$

$$\text{(iii)} \quad q_m \quad \text{(vi)}$$

$$M^* = q \left( \frac{\omega v}{\omega_m^r} \right)^{\frac{1}{3}}$$

$$\boxed{\frac{L}{v} = \frac{q}{\omega}} \Rightarrow$$

$$\boxed{\frac{\omega}{m\omega} = \frac{M}{1}} \Rightarrow$$

$$e_R = \omega L^* + v K^* + \omega_m M^* \quad \text{--- (iv)}$$

$$\Rightarrow e(q, \omega) = \omega \cdot q \cdot \left( \frac{\omega_m \cdot v}{\omega^r} \right)^{\frac{1}{3}} + v \cdot q \cdot \left( \frac{\omega_m \cdot \omega}{v^r} \right)^{\frac{1}{3}} + \omega_m \cdot q \cdot \left( \frac{\omega v}{\omega_m^r} \right)^{\frac{1}{3}}$$

We can derive Ac & Mc from it. (vii)  $\omega^0$

$$\frac{\partial}{\partial \omega} \left( \omega \cdot \frac{\omega}{m\omega} \cdot \dots \cdot \frac{\omega}{v} \right) =$$



$$c(q, w) = q (w_m \cdot w \cdot v)^{1/3} + q (w_m \cdot w \cdot v)^{1/3} + q (w \cdot v \cdot w_m)^{1/3}$$

$$= 3q (w_m \cdot w \cdot v)^{1/3}$$

$$AC_{LR} = \frac{c(q, w)}{q}$$

$$= 3 (w_m \cdot w \cdot v)^{1/3}$$

$$MC_{LR} = \frac{\partial c}{\partial q} = 3 (w_m \cdot w \cdot v)^{1/3}$$

In the long run  $AC = MC$  : (Proved) QED

So, zero economic profit. As  $[P = MC = AC]$   $[TR = TE]$

In the perfect competition.

Now if cost  $\uparrow$  economic inefficiency  $\rightarrow$  diseconomies of scale.

usually SR {

if cost  $\downarrow \rightarrow$  economies of scale.

LR — cost  $\rightarrow$  efficiency. constant returns to scale.

## Diminishing Returns / Economies of Scale

→ Diminishing (or increasing) returns

— short-run phenomenon

(What's happening with the marginal product of input).

Increasing returns — Marginal product is increasing.

Diminishing returns — Marginal product is decreasing.

→ Law of diminishing returns — as you add more of an input, eventually its marginal product decreases.

□ Constant Returns to scale (CRTS)

Double all inputs, output exactly doubles.

$f(K, L)$  is h.d. 1.



② Relationship between returns to scale and long-run average costs and marginal costs:

IRTS  $\leftrightarrow$  decreasing  $AC_{LR}$

CRTS  $\leftrightarrow$  constant  $AC_{LR}$

DRTS  $\leftrightarrow$  increasing  $AC_{LR}$

Diseconomies of scale  $\rightarrow$  Increasing  $AC_{LR}$

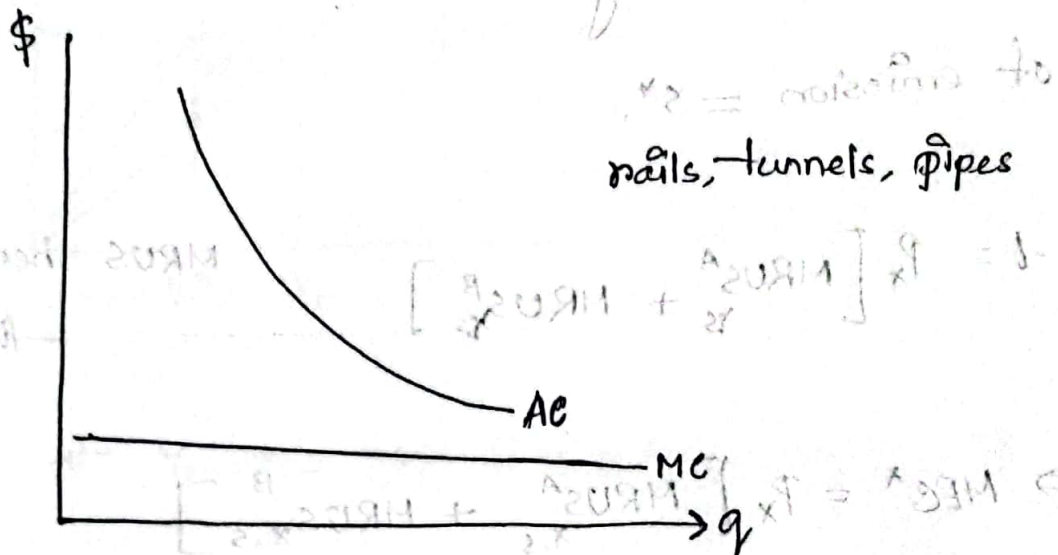
Economies of scale  $\rightarrow$  Decreasing  $AC_{LR}$

$$q = \frac{4KL}{K+L} \rightarrow h.d.1$$



Sources of (short-run) economies of scale

→ Big fixed or sunk cost (Natural monopolies)



Input specialization

Exploiting increasing returns

→ Advertising/Marketing

Eventually firms run into diseconomies of scale (i.e. then average costs start to increase).

→ Management constraints/co-ordination problems

→ Transportation costs

— Broking and other infrastructure  
Upward-sloping input - supply.

AC "Glossy"

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?

Advantages

2. Can include marginal cost as same graphs. (AC, AVC, MC are all measured in same units).

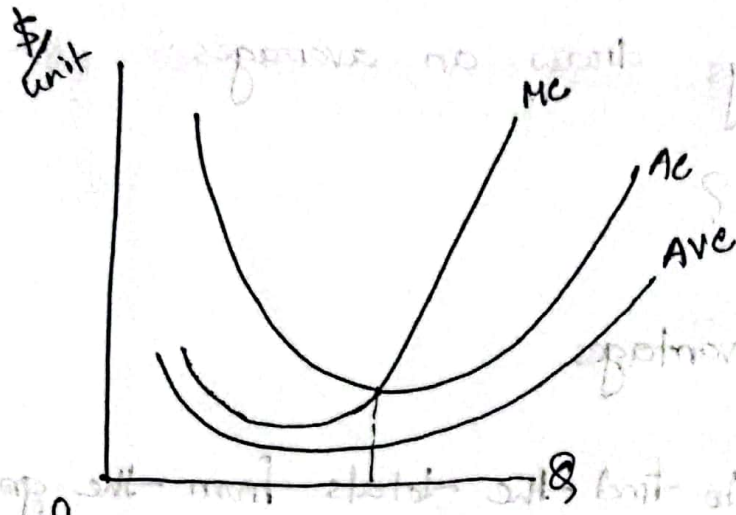
- deal about the

• *harmful*

scale — constant

2/3

"Classic" AC Graph:



Minimum Efficient ~~scale~~ <sup>sale</sup>: Lowest output level where minimum AC is reached.

$$TC = c(q, w)$$

$$AC = \frac{c(q, w)}{q}$$

let assume that AC is convex, so that minimum point can be obtained.

For,

$$\frac{\partial AC}{\partial q} = \frac{\frac{\partial c}{\partial q} \cdot q - c(q, w)}{q^2} = 0$$

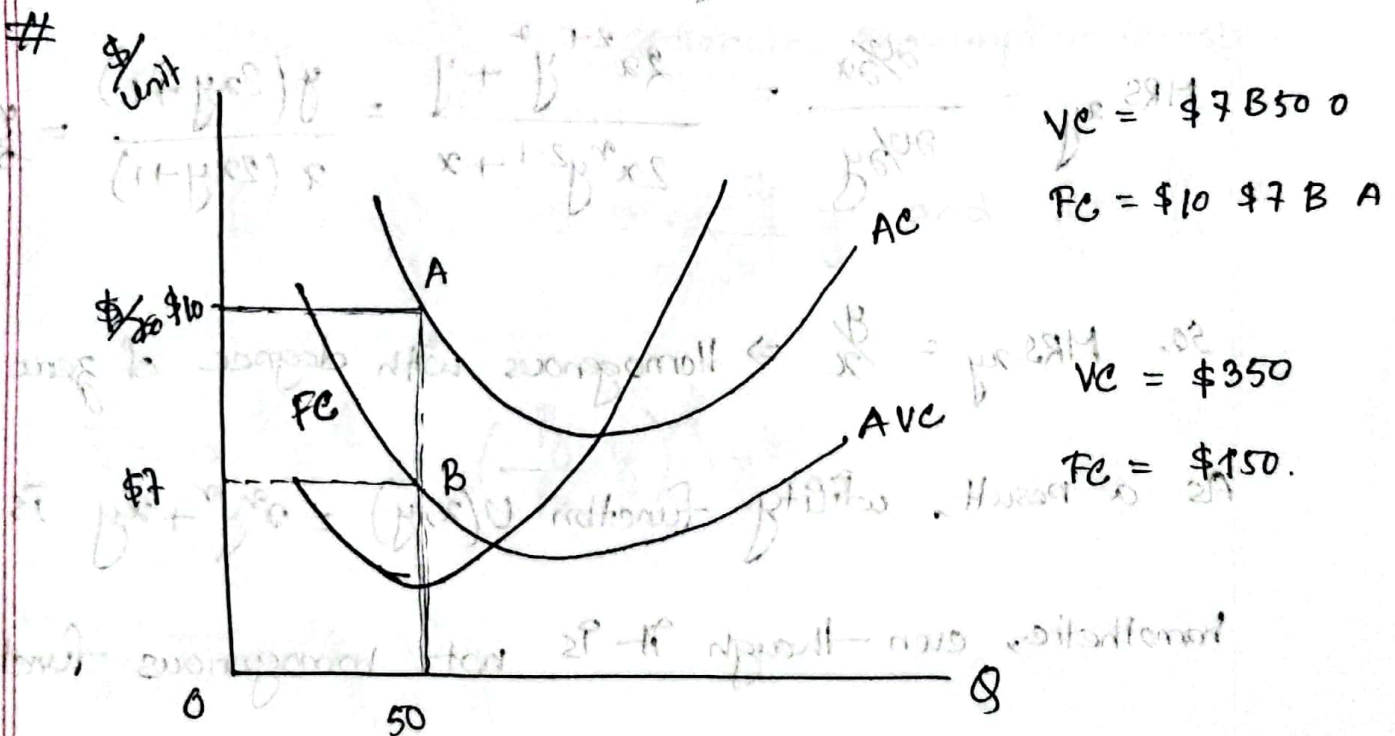
$$\Rightarrow \frac{\partial c}{\partial q} q - c(q, w) = 0$$



$$\Rightarrow \frac{\partial c}{\partial q} = \frac{e(q, w)}{q} \quad \left[ \text{As } \frac{\partial \pi}{\partial q} > 0, \text{ so we get} \right.$$

$$\Rightarrow MC = AC \quad \left. \text{the minimum.} \right]$$

[can find the same for  $AVC$  (?)]



$$\left[ \begin{aligned} AVC &= \frac{VC}{q} \end{aligned} \right.$$

$$\frac{\partial VC}{\partial q} = \frac{MC \cdot q - VC}{q^2}$$

$$= \frac{MC - AVC}{q} = 0 \quad \therefore MC = AVC$$

Ch Use cost curves to determine optimal quality

Goal of firm: Maximizing profit

$$\pi = TR - TC$$

Choose  $q$  to max  $\pi$

$$\underset{q}{\text{Max}} \quad \pi = TR - TC$$

$$\frac{\partial \pi}{\partial q} = MR - MC = 0$$

$\pi$ -maximizing rule: produce where  $MR = MC$

Assumed  $q$  uses continuous

→ We're pretty comfortable with MC.

MR is a little more complicated. It depends on industry structures.

3 broad ideas about MR

first one we did  $MR = P \left[ 1 + \frac{1}{E_{qp}} \right]$

$$TR = P(q) \cdot q$$

$$MR = \frac{\partial TR}{\partial q}$$

$$= \frac{\partial P}{\partial q} \cdot q + P$$

$$= P \left[ \frac{\partial P}{\partial q} \cdot \frac{q}{P} + \frac{P}{P} \right]$$

$$= P \left[ 1 + \frac{1}{E_{qp}} \right]$$

Monopoly or  
Monopolistic competitive  
market

Implicitly assumes no other firms react to your activity.

Second: Price taking - no ability to influence the price.

Can sell or buy all we want at the current price.



2 ways to think of this.

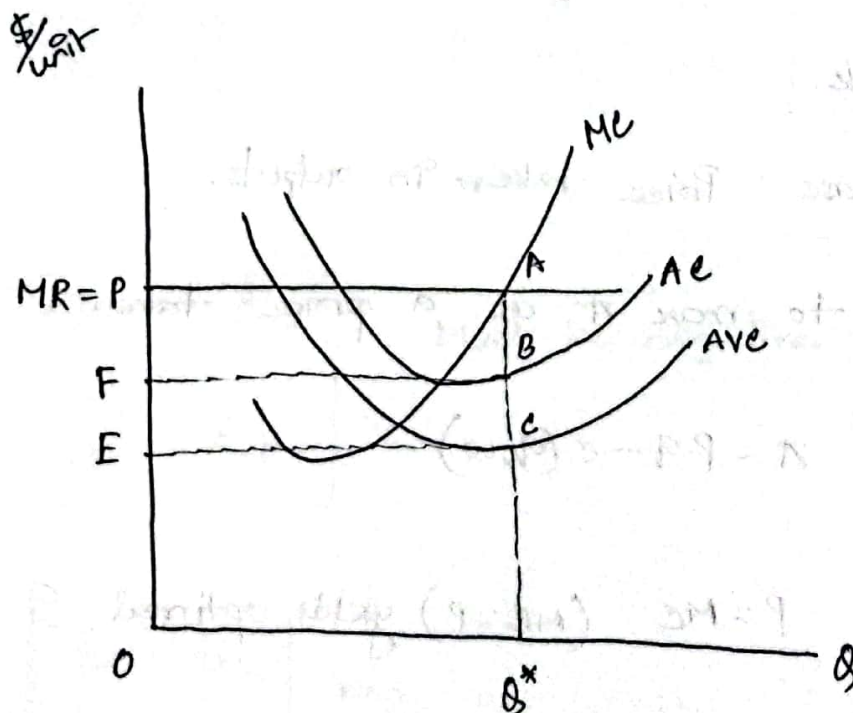
$$Eq, p = -\infty \text{ or } \frac{\partial P(q)}{\partial q} = 0$$

Perfect competitive

When a price taker,  $MR = P$

3rd:

The MR depends on the competitors, as one takes the decision after considering the decision of its rival.



$Q^*$  = profit maximizing output

$\pi = TR - TC = \text{area of box } BAPF > 0$

Operating profit,  $\pi_0 = TR - VC = \text{area of box } CAPE > 0$

$$[(\pi + Fe) = \pi_0]$$

↳ Short-run scenario

Why we need to know operating profit?

(a) Most important about input demand! a derived demand, derived from the demand for the output.

Math example:

Simplest case: Price takers in outputs.

2 ways to max  $\pi$  as a price taker

$$\text{Max}_q \quad \pi = P \cdot q - c(q, w)$$

yields  $P = MC$  ( $MC = P$ ) yields optimal  $q$ .

$$\pi = P \cdot f(K, L) - (wL + vk)$$

$$\frac{\partial \pi}{\partial L} = P \cdot \frac{\partial f}{\partial L} - w = 0 \quad \text{or} \quad P \cdot MP_L = w$$

$$\frac{\partial \pi}{\partial K} = P \cdot \frac{\partial f}{\partial K} - v = 0 \quad \text{or} \quad P \cdot MP_K = v$$

$P \cdot MP_i$  = Marginal revenue product of input  $i$  (when a price taker)

$$\frac{\partial \pi}{\partial L_i}$$

$$\text{SOC, } \frac{\partial^2 \pi}{\partial L^2} = P \cdot \frac{\partial MP_L}{\partial L} \leq 0 \quad (14)$$

$$\begin{bmatrix} P \cdot \frac{\partial MP_L}{\partial L} & P \cdot \frac{\partial MP_L}{\partial K} \\ P \cdot \frac{\partial MP_K}{\partial L} & P \cdot \frac{\partial MP_K}{\partial K} \end{bmatrix}$$

$$\begin{bmatrix} \pi_{LL} & \pi_{LK} \\ \pi_{LK} & \pi_{KK} \end{bmatrix}$$

Must be negative semidefinite

$$P \begin{bmatrix} f_{LL} & f_{KL} \\ f_{KL} & f_{KK} \end{bmatrix}$$

neg semidefinite

$$f_{LL} < 0, f_{KK} < 0 \quad \text{diminishing MP.}$$

$$f_{LL} \cdot f_{KK} > f_{KL}^2$$



Negative semidefinite  $\rightarrow$  diminishing returns  
profit max

Example: Price taker in output

$$Q = (KL)^{\frac{1}{3}}$$

Max  
L, K

$$\pi = P(KL)^{\frac{1}{3}} - (wL + vK)$$

$$\frac{\partial \pi}{\partial L} = \frac{1}{3} P \left( \frac{K}{L^2} \right)^{\frac{1}{3}} - w = 0 \quad \text{--- ①}$$

$$\frac{\partial \pi}{\partial K} = \frac{1}{3} P \left( \frac{L}{K^2} \right)^{\frac{1}{3}} - v = 0 \quad \text{--- ②}$$

If take ratio,  $\frac{K}{L} = \frac{w}{v}$

$$K = \frac{w}{v} \cdot L$$

Substituting in ①:

$$\frac{1}{3} P \left( \frac{\frac{w}{v} \cdot L}{L^2} \right)^{\frac{1}{3}} = w$$

$$\Rightarrow P \left[ \left( \frac{\frac{wL}{v}}{L^2} \right)^{\frac{1}{3}} \right] = 3w$$

$$\Rightarrow P \left[ \frac{w}{v} \right]^{\frac{1}{3}} L^{-\frac{1}{3}} = 3w \text{ (given to find } L)$$

$$\Rightarrow \frac{P}{3w} \left( \frac{w}{v} \right)^{\frac{1}{3}} = L^{\frac{1}{3}}$$

$$\Rightarrow \frac{P^3}{27w^3} \cdot \frac{w}{v} = L$$

$$L^*(P, w) = \frac{P^3}{27w^3 v}$$

$$K = \frac{w}{v} \cdot \frac{P^3}{27w^3 v}$$

$$K^*(P, w) = \frac{P^3}{27wv^2}$$

$$\# \quad q = (KL)^{\frac{1}{3}}$$

$$= (\theta K \theta L)^{\frac{1}{3}} = \theta^{\frac{2}{3}} (KL)^{\frac{1}{3}} ; \frac{2}{3} < 1 \text{ decreasing returns to scale}$$

$$q = \frac{4KL}{K+L} ; \text{ constant returns to scale.}$$



Elasticity of substitution: from  $q = \frac{4KL}{K+L}$

$$\sigma = \frac{\% \Delta \frac{K}{L}}{\% \Delta MRTS} \quad \left[ = \frac{d \log \frac{K}{L}}{d \log MRTS} \right]$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{\omega}{v}$$

$$\sigma = \frac{(\Delta \frac{K}{L}) / (\frac{K}{L})}{(\Delta MRTS) / MRTS}$$

$$= \frac{\Delta \frac{K}{L}}{\frac{K}{L}} \cdot \frac{MRTS}{\Delta MRTS}$$

$$= \frac{\Delta \frac{K}{L}}{\Delta MRTS} \cdot \frac{MRTS}{\frac{K}{L}}$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{4K(K+L) - 4KL}{(K+L)^2} \cdot \frac{4L(K+L) - 4KL}{(K+L)^2}$$

$$= \frac{K^2}{L^2}$$



$$MRTS = \left(\frac{K}{L}\right)^2$$

$$\log MRTS = 2 \log \left(\frac{K}{L}\right)$$

$$\Rightarrow \log \left(\frac{K}{L}\right) = \frac{1}{2} \log(MRTS)$$

$$\sigma = \frac{d \log \left(\frac{K}{L}\right)}{d \log(MRTS)} = \frac{1}{2} \quad \left[ \begin{array}{l} \text{what does it mean?} \\ \text{perfect substitute} \\ \text{when?} \end{array} \right]$$

10 A price-taking firm's cost function is estimated to be.

$$C(q) = 100 + 16q - 4q^2 + \left(\frac{1}{3}\right)q^3$$

- Is this a short-run or long-run cost function?
- You have a production run of 50 for this period. How much will it cost to produce? If price is 500, what is your operating profit?

2 effect when an input price change

1. Relative price of the other inputs change.

— substitute toward the relatively cheaper inputs

Substitution effect: Bigger the elasticity of substitution  $\sigma$  the bigger the substitution effect and therefore the more elastic is input demand.

2. Inputs price goes up. My costs also increase and thus eventually, my optimal level of output will decrease. This changes amount of all inputs used (decreases)

Called the output effect.

↳ Depends on the rate of output change.

↳  $\sigma$  is higher if the availability  $\uparrow$  meaning price  $\downarrow$ .

Elastic input demand  $\rightarrow$  availability higher.



Output effect, 2 parts

— How much your costs increased

→ Bigger the share of cost of the input, more

costs increase → bigger the output effect

$\left[ \frac{\% \text{ change in output}}{\% \text{ change in input}} \right] \rightarrow$  more elastic is input demand

— Elasticity of the demand for what you are making

more elastic → bigger output effect → More elastic is input demand.

Availability ↑ → elasticity of substitution ↑

not just elasticity

Here we consider input elasticity =  $\frac{\% \Delta \text{ in output}}{\% \Delta \text{ in input}}$

$$= \frac{\% \Delta Q}{\% \Delta P} = \frac{\% \Delta Q}{\% \Delta P} = \frac{\% \Delta Q}{\% \Delta P}$$



$$C(q) = 100 + 16q - 4q^2 + \left(\frac{1}{3}\right)q^3$$

a. This is a short-run cost function because there is a fixed cost of \$100.

When,  $q = 0$ , then the total cost is \$100.

b. When  $q = 50$ ,

$$C(50) = 100 + 16(50) - 4(50)^2 + \frac{1}{3}(50)^3$$

$$= 100 + 800 - 10,000 + 41,666.67$$

$$= \$32,566.67$$

Operating profit  $\pi_0 = TR - VC$

$$\text{Here, } TR = P \times q = 500(50)$$

$$= \$25,000$$

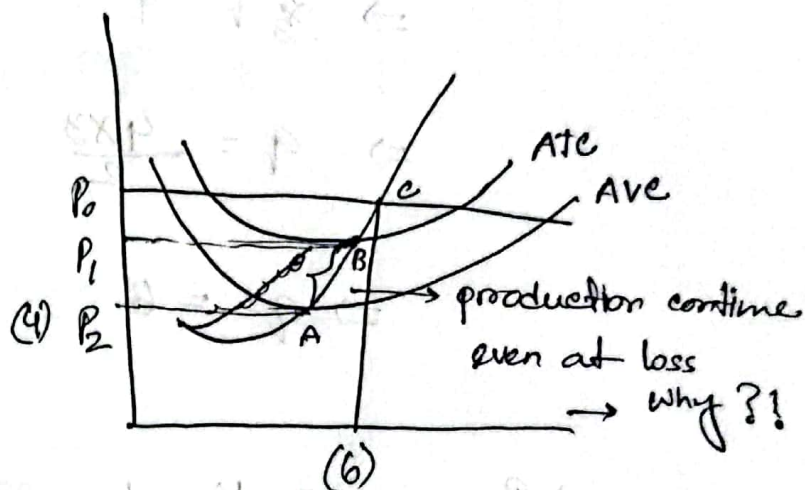
$$VC = 16q - 4q^2 + \frac{1}{3}q^3$$

$$= 16(50) - 4(50)^2 + \frac{(50)^3}{3} = 32,466.67$$

$$\pi_0 = 25000 - 32,466.67$$

$$= \text{Rs } 7,466.67 \text{ (Loss)}$$

Shut down price



c is break even price

A is shut down price.

c. What is the shut down price?

$$\Rightarrow \text{Total variable cost (TVC)} = 16q - 4q^2 + \frac{1}{3}q^3$$

Average variable cost (Ave) =  $\frac{TVC}{q} = 16 - 4q + \frac{1}{3}q^2$

At shutdown point, AVC is minimum

that  $q_5$   $P_{\text{shutdown}} = \min \text{AVE}$

FOC,

$$\frac{\partial \text{Ave}}{\partial q} = -4 + \frac{2}{3}q = 0$$

$$\Rightarrow \frac{2}{3}q = 4$$

$$\Rightarrow q = \frac{4 \times 3}{2}$$

$$\Rightarrow q = 6$$

$$\therefore P_{\text{shutdown}} = 16 - 4q + \frac{q^2}{3}$$

$$= 16 - 4(6) + \frac{6^2}{3}$$

$$= 16 - 24 + 12$$

$$= 4$$

d. The market price is \$36. What is the profit maximizing quantity? What is your short-run profit.



⇒ Profit maximizing condition,  $MR = MC = P$

$$TC = 100 + 16q + 4q^2 + \frac{1}{3}q^3$$

$$MC = \frac{\partial TC}{\partial q} = 16 + 8q + q^2 = P$$

⇒

Since

$$P = 36$$

$$16 + 8q + q^2 = 36$$

$$\Rightarrow q^2 + 8q - 20 = 0$$

$$\Rightarrow q^2 - 10q + 2q - 20 = 0$$

$$\Rightarrow q(q-10) + 2(q-10) = 0$$

$$\Rightarrow (q+2)(q-10) = 0$$

$$\Rightarrow q = -2 \quad \text{or} \quad q = 10$$

Since  $q \neq -2$   $\therefore q = 10$ .

quantity cannot be

negative. So,  $q = 10$ .

$$\begin{aligned}
 \text{Short run profit, } \pi_{sr} &= p \cdot q - c(q) \\
 &= 36(10) - \left[ 100 + 16(10) - 4(10)^2 + \frac{(10)^3}{3} \right] \\
 &= 360 - [100 + 160 - 400 + 333.33]
 \end{aligned}$$

$$q = 166.67 \quad \frac{516}{3} = 172$$

Chapter - 11 Exercise

Chapter - 9, 10

$$28 = 9$$

$$28 = r_p + r_{28} - 21$$

$$0 = 28 - r_p - 21 \Rightarrow$$

$$0 = 28 - r_p + 21 - r_p \Rightarrow$$

$$0 = (28 - r_p) + (21 - r_p) \Rightarrow$$

$$0 = (28 - r_p) (21 - r_p) \Rightarrow$$

$$28 - r_p = 0$$

$$21 - r_p = 0$$

$$r_p = 28$$